Non-Stoichiometry Defects and Density Decrease in FeO

Original statement · Problems T2 defects.pdf

5. In a specimen of iron oxide (II), FeO*,* 10% *of the* Fe^{2+} *ions are substituted by* Fe^{3+} *ions. In order to maintain the crystal electro neutrality, a fraction of the cationic positions are vacancies. Data:* $M(\mathrm{O}) = 16\ \mathrm{g/mol}$*;* $M(\mathrm{Fe}) = 55.8\ \mathrm{g/mol}$*. a) Calculate the fraction of cationic vacancies. (Solution:* $0.033\ \mathrm{mol\ vacancies/mol\ Fe^{2+}}$*) b) Assuming that the lattice parameter of the perfect and imperfect crystal is the same, calculate how much the density decreases, in* $\%$*, with respect to the perfect crystal. (Solution:* -2.56%*)*

Solution: $0.033\ \mathrm{mol\ vacancies/mol\ Fe^{2+}}$

PHASE 1: HYPOTHESIS & PARAMETERS

Chemical and Physical Context:

Iron(II) oxide (FeO, wüstite) crystallises in the rock-salt structure (NaCl, FCC) with a close-packed anionic sublattice of O^{2-} and a cationic iron sublattice. In oxidising atmospheres, part of the Fe^{2+} ions are oxidised to Fe^{3+}.

Input Data:

- Substitution: 10% of the original Fe^{2+} positions are replaced following $3\,\mathrm{Fe}^{2+} \rightarrow 2\,\mathrm{Fe}^{3+} + 1\,V''_{\mathrm{Fe}}$.
- Atomic molar masses:
- $M_{\mathrm{Fe}} = 55.8\ \mathrm{g/mol}$
- $M_{\mathrm{O}} = 16.0\ \mathrm{g/mol}$
- Molar mass of the ideal stoichiometric FeO unit:

$$M_{\mathrm{ideal}} = 55.8 + 16.0 = 71.8\ \mathrm{g/mol}$$

- Geometric hypothesis: The lattice parameter a remains unchanged between the perfect and the defective crystal ($V_C^{\mathrm{real}} = V_C^{\mathrm{ideal}}$).

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Electroneutrality Principle [Session 4 Slide 22]:**

Each Fe^{3+} ion contributes an excess charge of $+1$ relative to the normal Fe^{2+} position. To preserve overall electrical neutrality:

$$3\,Fe^{2+} \longrightarrow 2\,Fe^{3+} + 1\,V''_{Fe}$$

For every $2\,Fe^{3+}$ ions present, exactly 1 **cationic iron vacancy** must be created.

2. **Cationic Vacancy Fraction:**

Basis: 1 mol of cation sites of the perfect crystal (1 mol of Fe^{2+}, with 1 mol of O^{2-}, total anionic charge -2).

The replacement of 10% of the Fe^{2+} positions follows the reaction $3\,Fe^{2+} \rightarrow 2\,Fe^{3+} + 1\,V''_{Fe}$, so every 3 replaced ions give 1 vacancy; this is the single rule used in part a).

3. **Percentage Density Change:**

Since the lattice volume V_C is constant, the density is strictly proportional to the effective molar mass per formula unit:

$$\frac{\Delta\rho}{\rho_0} = \frac{M_{\text{real}} - M_{\text{ideal}}}{M_{\text{ideal}}} \times 100\%$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Part a: Cationic Vacancy Fraction

- Basis: 1 mol of cation sites of the perfect crystal, i.e. 1 mol of Fe^{2+} and 1 mol of O^{2-} (the unit used in the official answer, "mol vacancies/mol Fe^{2+}").
- Reading of the statement: 10% of the original Fe^{2+} positions are replaced, i.e. 0.10 mol of the original Fe^{2+} are involved in the reaction

$$3\,Fe^{2+} \longrightarrow 2\,Fe^{3+} + 1\,V''_{Fe}$$

- Check of the reaction: sites $3 = 2 + 1$; charge $3 \times (+2) = 2 \times (+3) + 0 = +6$ (the vacancy carries no ion).
- Every 3 replaced Fe^{2+} produce 1 vacancy, hence for 0.10 mol of replaced Fe^{2+}:

$$n_{\text{vac}} = \frac{0.10}{3} = 0.03333 \text{ mol} \approx \mathbf{0.033} \text{ mol vacancies / mol } \mathbf{Fe^{2+}}$$

- Resulting populations (per mol of original cation sites): $n_{Fe^{3+}} = 2\,n_{vac} = 0.06667$; $n_{Fe^{2+}} = 1 - 0.10 = 0.90$; $n_{vac} = 0.03333$.
- Checks: sites $0.90 + 0.06667 + 0.03333 = 1.000$; charge $0.90 \times 2 + 0.06667 \times 3 = 1.800 + 0.200 = +2.000$, equal in magnitude to the anionic charge of 1 mol of O^{2-} (-2).

Value coincides with the official answer 0.033.

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2. Part b: Percentage Density Change

- Fe atoms per formula unit (per O): $0.90 + 0.06667 = 0.96667$, i.e. the crystal is $Fe_{0.9667}O$ ($= 1 - 0.03333$).
- **Effective molar mass of the defective crystal (per formula unit):**

$$M_{real} = 0.96667 \times 55.8 + 16.0 = 53.940 + 16.0 = 69.940 \text{ g/mol}$$

- **Molar mass of the perfect crystal:** $M_{ideal} = 55.8 + 16.0 = 71.80$ g/mol.
- **Density change** (same cell volume, so $\rho \propto M$ per formula unit):

$$\frac{\Delta\rho}{\rho_0} = \frac{M_{real} - M_{ideal}}{M_{ideal}} \times 100\% = \frac{69.940 - 71.80}{71.80} \times 100\% = \frac{-1.860}{71.80} \times 100\% = \mathbf{-2.59\%} \approx \mathbf{-2.6\%}$$

Equivalently, the mass lost is only the vacancies, $\Delta M = -n_{vac} M_{Fe} = -0.03333 \times 55.8 = -1.860$ g per 71.80 g of perfect FeO.

Discrepancy with the official solution. The official key gives -2.56%; the value obtained from the stated data ($M_{Fe} = 55.8$, $M_O = 16$) is -2.59%. The difference (0.03 percentage points) cannot be traced to any rounding of the given constants, so the official figure is not reproduced and no data were altered. A different reading of the statement (10% of the Fe ions of the final crystal being Fe^{3+}) would give 0.0476 vacancies per formula unit and -3.7%, which matches neither official value and is therefore not used.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **The Wüstite Phenomenon:** FeO practically never exists with exact $1:1$ stoichiometry at room temperature; it always appears as an iron-deficient non-stoichiometric phase $Fe_{1-x}O$ ($0.05 \leq x \leq 0.15$). The presence of cationic vacancies V''_{Fe} reduces its macroscopic density (in this problem, $\approx -2.6\%$) and gives the material p-type semiconductor properties.

Study notes written from the official course material (Problems T2_CrystStruct.pdf; Problems T2 defects.pdf).
Online version: subjects/aerospace-materials-1/problemas/topic-2-problems.html